Complex numbers: the geometric point of view

Lecture and worked examples

Objectives

Represent a complex number by a point or a vector; determine a modulus and an argument; move from the algebraic form to the trigonometric form and back; interpret a modulus as a distance.

Introductory activity

The plane has an orthonormal frame $(O, \vec{u}, \vec{v})$. To a complex number $x + iy$ we associate the point with coordinates (x, y).

Introductory activity (2)

1. Plot the points A, B, C associated with $a = 2 + i$, $b = i\,a$ and $c = (1 + i)\,a$. Give b and c in algebraic form.
2. Compute the distances OA, OB and OC.
3. Compute the scalar product $\overrightarrow{OA} \cdot \overrightarrow{OB}$. What can be said about the lines (OA) and (OB)?
4. Compute $\cos\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$ with the scalar product, then the sign of the determinant $\det\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$. Deduce a measure of the oriented angle $\left(\overrightarrow{OA}, \overrightarrow{OC}\right)$.

Introductory activity (3)

5. Start again with $a = -1 + 3i$ (or a point of your choice). State a conjecture about the geometric effect of a multiplication by i, then by $1 + i$.
6. Which quantities attached to a complex number would have to be defined to state this conjecture in general?

Introductory activity (4)

Observing is not proving: the properties conjectured here are established in the course.

Complex plane

The plane has a direct orthonormal frame $(O, \vec{u}, \vec{v})$. To the complex number $z = a + ib$ (a, b real) we associate the point $M(a, b)$, called the *image* of z, and the vector $\vec{w} \begin{pmatrix} a \\ b \end{pmatrix}$. We say that z is the *affix* of M and of $\vec{w}$. The plane with this correspondence is called the *complex plane*.

Complex plane

If A and B have affixes z_A and z_B, the vector $\overrightarrow{AB}$ has affix $z_B - z_A$ and the midpoint of $[AB]$ has affix $\dfrac{z_A + z_B}{2}$. The affix of $\overrightarrow{w} + \overrightarrow{w'}$ is the sum of the affixes, that of $k\overrightarrow{w}$ (k real) is k times the affix of $\overrightarrow{w}$.

The image of $\bar{z}$ is the reflection of the image of z in the horizontal axis; the image of $-z$ is its reflection in O.

Modulus

The *modulus* of $z = a + ib$ is the nonnegative real number $|z| = \sqrt{a^2 + b^2}$. If M is the image of z, $|z| = OM$. For a real number, the modulus is the absolute value.

Modulus

For every complex number z, $|z|^2 = z\bar{z}$, and $|\bar{z}| = |-z| = |z|$. If A and B have affixes z_A and z_B, then $AB = |z_B - z_A|$.

Modulus

With $z = a + ib$, $z\bar{z} = a^2 + b^2 = |z|^2$ (chapter 1). The distance AB is the norm of the vector $\overrightarrow{AB}$, whose affix is $z_B - z_A$.

Modulus

For all complex numbers z, z' and every natural number n:

$$|zz'| = |z|\,|z'|\,, \qquad |z^n| = |z|^n\,, \qquad \text{and, if } z' \neq 0, \quad \left|\frac{1}{z'}\right| = \frac{1}{|z'|}, \quad \left|\frac{z}{z'}\right| = \frac{|z|}{|z'|}.$$

Modulus

Product. $|zz'|^2 = zz'\,\overline{zz'} = z\bar{z}\,z'\overline{z'} = |z|^2\,|z'|^2$. Moduli are nonnegative, so $|zz'| = |z|\,|z'|$.

Power. By induction: $\left|z^0\right| = 1 = |z|^0$, and if $|z^n| = |z|^n$, then $\left|z^{n+1}\right| = |z^n|\,|z| = |z|^{n+1}$.

Inverse and quotient. $|z'| \times \left|\frac{1}{z'}\right| = \left|z' \times \frac{1}{z'}\right| = 1$, then $\left|\frac{z}{z'}\right| = |z|\left|\frac{1}{z'}\right|$.

Modulus

For all complex numbers z and z', $|z + z'| \leq |z| + |z'|$. The problem of the chapter gives a proof and the case of equality.

Modulus

We write $\mathbb{U}$ for the set of complex numbers of modulus 1. Their images form the circle with centre O and radius 1.

Modulus

If z and z' belong to $\mathbb{U}$, then $zz' \in \mathbb{U}$ and $\dfrac{1}{z} \in \mathbb{U}$; moreover $\dfrac{1}{z} = \overline{z}$.

Modulus

$|zz'| = 1 \times 1 = 1$ and $\left|\frac{1}{z}\right| = \frac{1}{1} = 1$. Finally $z\bar{z} = |z|^2 = 1$, so $\bar{z} = \frac{1}{z}$.

Modulus

Let A and B have affixes a and b, and M have affix z. The equality $|z-a| = r$ $(r > 0)$ means $AM = r$: M describes the circle with centre A and radius r. The equality $|z-a| = |z-b|$ $(a \neq b)$ means $AM = BM$: M describes the perpendicular bisector of $[AB]$.

Argument and trigonometric form

Let z be a nonzero complex number with image M. An *argument* of z is any measure, in radians, of the oriented angle $\left(\overrightarrow{u}, \overrightarrow{OM}\right)$. We write $\arg z$ for one of them; the others differ from it by a multiple of 2π, which we write $\arg z \equiv \theta \pmod{2\pi}$. The number 0 has no argument.

Argument and trigonometric form

Every nonzero complex number $z = a + ib$ can be written $z = r(\cos\theta + i\sin\theta)$ with $r = |z| > 0$ and $\theta \equiv \arg z \pmod{2\pi}$; then

$$\cos\theta = \frac{a}{r}, \qquad \sin\theta = \frac{b}{r}.$$

Conversely, if $z = r(\cos\theta + i\sin\theta)$ with $r > 0$, then $|z| = r$ and θ is an argument of z. Two trigonometric forms $r(\cos\theta + i\sin\theta)$ and $r'(\cos\theta' + i\sin\theta')$ are equal if and only if $r = r'$ and $\theta \equiv \theta' \pmod{2\pi}$.

Argument and trigonometric form

For $z \neq 0$: $\arg \bar{z} \equiv -\arg z$ and $\arg(-z) \equiv \arg z + \pi \pmod{2\pi}$. A positive real number has argument 0, a negative real number π; i has argument $\frac{\pi}{2}$.

Argument and trigonometric form

Compute $r = |z|$, then $\frac{a}{r}$ and $\frac{b}{r}$, and recognise the cosine and sine of a standard angle.

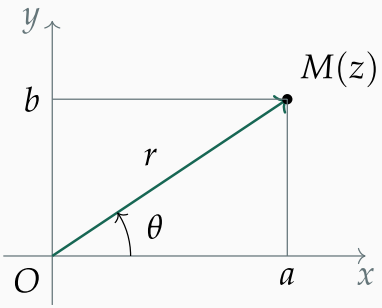
Argument and trigonometric form

$1 + i\sqrt{3}$: $r = 2$, $\cos\theta = \dfrac{1}{2}$, $\sin\theta = \dfrac{\sqrt{3}}{2}$, so $1 + i\sqrt{3} = 2\left(\cos\dfrac{\pi}{3} + i\sin\dfrac{\pi}{3}\right)$.

$-1 + i = \sqrt{2}\left(\cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4}\right)$; $-3 = 3(\cos\pi + i\sin\pi)$; $-2i = 2\left(\cos\left(-\dfrac{\pi}{2}\right) + i\sin\left(-\dfrac{\pi}{2}\right)\right)$.

Argument and trigonometric form

$-2\left(\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}\right)$ is not a trigonometric form, because $-2<0$. We write
$-2\left(\cos\frac{\pi}{5}+i\sin\frac{\pi}{5}\right)=2\left(\cos\frac{6\pi}{5}+i\sin\frac{6\pi}{5}\right)$.



Argument and trigonometric form

$|z| = OM$, $|z_B - z_A| = AB$, $|z|^2 = z\overline{z}$; the modulus of a product is the product of the moduli; $z = r(\cos\theta + i\sin\theta)$ with $r > 0$; an argument is only defined up to a multiple of 2π.